JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 100, NO. B8, PAGES 15,009–15,023, AUGUST 10, 1995

Remagnetization of the folded Belden Formation, northwest Colorado

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Abstract. Paleomagnetic and geochemical results are used to evaluate the origin of an apparent synfolding magnetization in Belden Formation (Pennsylvanian) limestones on an asymmetric fold in Colorado. The natural remanent magnetizations contain three components: a modern viscous magnetization removed at low temperatures, a reversed Tertiary magnetization (declination (D)=186.7°, inclination (I)=-54.6°) of thermoviscous origin removed at intermediate temperatures, and an apparently synfolding Late Cretaceous/early Tertiary characteristic magnetization (D=338.7°E, I=62.5°, precision parameter k=54.9, radius of 95% probability cone $\propto_{95}$=5.9°, 65% unfolding) removed by 580°C. Hysteresis properties suggest that the characteristic magnetization resides in single-domain magnetite. Low burial temperatures and the presence of authigenic magnetite are consistent with a chemical origin for this magnetization. The $^{87}Sr/^{86}Sr$ values (0.7088-0.7100) show the limestones to be radiogenic, with ^{87}Sr enriched halos around calcite-cemented tectonic fractures. There is no control on the distribution and intensity of the magnetization by the veins or structural position (e.g., highly mineralized versus less mineralized rock). Lithology, rather than fluid alteration, controls the magnetization. Fecal pellets contain pyrite, with alteration rims identified as magnetite by scanning transmission electron microscope X ray diffraction patterns. Selective demagnetization has established that the fecal pellets carry the characteristic magnetization. It is not possible to unequivocally establish whether the characteristic magnetization is synfolding or prefolding. A synfolding magnetization caused by the syndeformational mineralizing fluids, however, can be ruled out. Structural modification of a prefolding component and a synfolding piezoremanent magnetization cannot be ruled out but are not considered likely mechanisms of remanence acquisition. Overprinting of a prefolding chemical magnetization by a postfolding viscous magnetization residing in multidomain grains is a likely explanation for the synfolding test result. The replacement of pyrite by magnetite is interpreted as prefolding in origin, perhaps triggered by an oxidizing fluid or by diagenesis of hydrocarbons.

Introduction

Fluids expelled from orogenic belts have received considerable attention as a potential agent of chemical remagnetization [e.g., *Miller and Kent*, 1988; *McCabe and Elmore*, 1989; *McCabe et al.*, 1989; *Bethke and Marshak*, 1990; Lu et al., 1991; *Oliver*, 1992; *Saffer and McCabe*, 1992; *Elmore et al.*, 1993a]. The link between chemical remanent magnetization (CRM) and orogenic fluids is based largely on a spatial association with orogenic belts and a temporal association with periods of orogeny [*McCabe and Elmore*, 1989]. Many of these CRMs are synfolding (e.g., *McCabe et al.* [1983], *Scotese et al.* [1985], *Bachtadse et al.* [1987], *Kodama* [1988], *Elmore et al.* [1993a], and many others) which has also been cited as evidence for a connection with orogenic fluids. In this model, orogenesis mobilizes fluids and provides conduits, such as fractures, for the fluids to more easily enter some rock types and cause chemical changes.

Whereas orogenic fluids are a possible agent of remagnetization, tests of the orogenic fluid model are rare. Direct links between CRMs, inferred fluids, and the magnetic carriers need to be made if secondary magnetizations are to provide useful insights into the diagenetic history of remagnetized rocks. The Pennsylvanian Belden Formation on a large asymmetric anticline in northwestern Colorado provides an opportunity to test the orogenic fluid model and to link an apparent synfolding Late Cretaceous-early Tertiary magnetization residing in magnetite to the processes which caused the magnetization using geology, geochemistry, and paleomagnetism.

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Paper number 95JB00045.
0148-0227/95/95JB-00045$05.00

Geologic Setting of the Study

The study area is located at the junction of Sweetwater Creek and the Colorado River (Figure 1) on the eastern flank

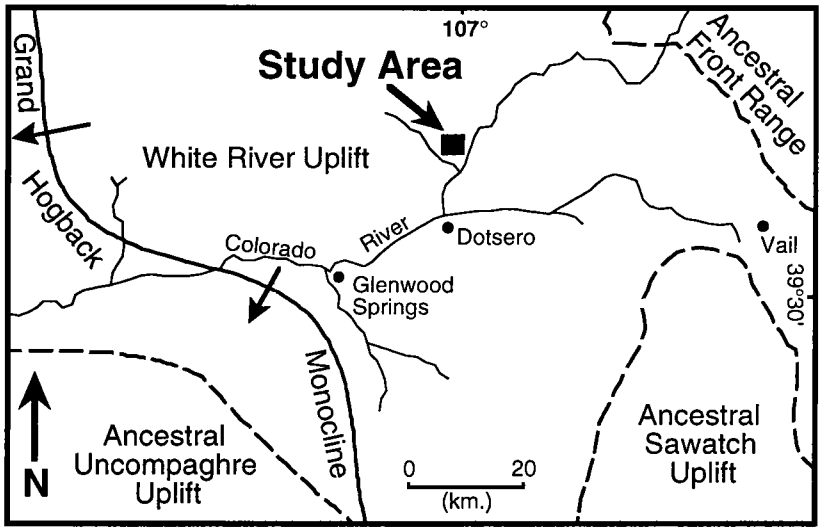


Figure 1. Location diagram for the study area at the intersection of Sweetwater Creek and the Colorado River, northwest Colorado (107.7°W longitude, 39.8°N latitude).

of the White River Uplift. The Belden Formation is exposed in the Sweetwater locality on a large asymmetric anticline (Figure 2). The lower contact of the Belden Formation is exposed on the shallow flank of the anticline and is placed at the first dark gray limestone or shale above the red and yellow clays of the Molas Formation or at the top of the light gray massive Leadville Limestone [*Murray*, 1958]. The contact with the overlying Minturn Formation is arbitrary, with predominantly shale and limestone in the Belden Formation and predominantly sandstone in the Minturn Formation [*Murray*, 1958].

The asymmetric fold at Sweetwater Creek dips as much as 80° on the steep western flank but only 15° on the eastern limb. A subsidiary fold in the hinge area resulted from shale flowage into the hinge. The limestones on the steep limb of the fold contain many fractures which formed as a result of deformation. Some of these fractures are entirely cemented by calcite, some of which is twinned, and many fracture fillings are deformed. The exposed Belden Formation is 150 m thick and consists of dark shales and fossiliferous limestones with occasional thin arkosic sandstones. Overall, the Belden Formation coarsens upward and is composed of several shoaling upward sequences, each starting with marine shale followed by carbonate mudstones, wackestones, packstones/grainstones, and, in some cases, capped by an arkosic sandstone. The lower part of the Belden Formation is not exposed on the steep flank, and some parts of the section are covered on the shallow flank. Nevertheless, certain beds can be traced across the entire structure.

Belden Formation rocks in the Sweetwater area reached a

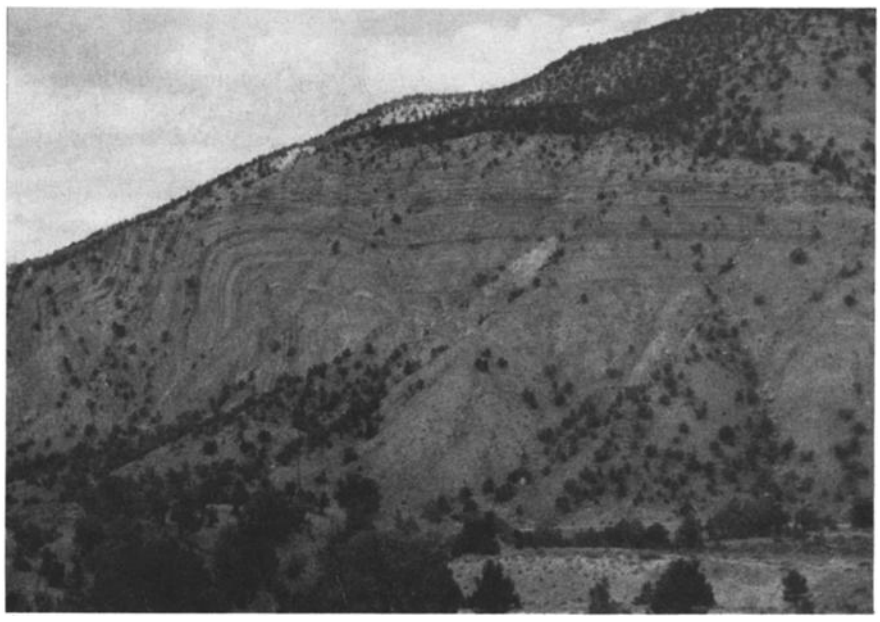


Figure 2. Hinge of the fold at Sweetwater Creek (viewed from the north). Trees 2-3m high are scale.

maximum burial temperature of approximately 150°C by the Late Cretaceous based on vitrinite reflectance and thermal maturity modeling assuming a variable geothermal gradient [*Nuccio et al.*, 1989]. The modeled temperatures exceed those necessary to generate petroleum [*Tissot and Welte*, 1984]. The White River Uplift began to rise at about 67 Ma [*Nuccio et al.*, 1989]. The fold at Sweetwater is Laramide in age which is consistent with the age of other structures in the region [*Davis*, 1978].

Sampling and Processing

The complete range of carbonate and clastic lithologies exposed in the anticline were sampled. A portable gas-powered drill was used to sample the rocks, and 2.5-cm-diameter cores were oriented in place using a magnetic compass and a clinometer. The oriented cores were then cut to a standard length of 2.2 cm in the laboratory.

Approximately 200 oriented core samples were collected from 25 sites located along the Sweetwater anticline from three general areas: the highly mineralized and fractured hinge zone; the highly fractured and mineralized steep flank (westerly dipping flank); and the less fractured, generally nonmineralized beds on the easterly, shallowly dipping limb. Some samples were collected from the same beds in all three areas. Sites designated SW11A (steep flank), SW11B (hinge area), and SW11C (shallow flank) are all from the same packstone-grainstone. Sites AM2 (steep), SAM2 (shallow), TAM2 (hinge) and TAMA2 (hinge) are all from the same wackestone bed; and AM1 (steep), SAMA1 (shallow), and TAM1 (hinge) are from the underlying carbonate mudstone (sites are sorted by structural position in Table 1a). Sites SW2S, SW3 and SW10 are from different mudstone beds on the steep flank, shallow flank and hinge positions, respectively. SWC samples are from a bed near the Minturn Formation contact, stratigraphically higher than the other beds sampled.

Most magnetic measurements were made using a United Scientific three-axis cryogenic magnetometer, sensitive to 10^{-6} A/m, that is located in a magnetically shielded laboratory which reduces the ambient magnetic field to less than 300 nT. Magnetic directions were plotted as orthogonal projections [*Zijderveld*, 1967], and best fit lines were calculated for linear demagnetization trajectories using principal component analysis [*Kirschvink*, 1980]. Paleomagnetic techniques used include thermal and alternating field (AF) demagnetization and selective destructive demagnetization (SDD), by which magnetic grains were removed selectively. For thermal demagnetization of the natural remanent magnetization (NRM) the specimens were heated at increasing temperatures (12 steps up to 580°C) in a Schonstedt thermal specimen demagnetizer (model TSD-1) oven for 30 min. AF demagnetization was performed using a two-axis tumbling unit located inside a variable field electromagnet. The specimens were subjected to increasing AF strengths (10 steps from natural remanent magnetization up to 125 mT) while rotating about two axes.

Acquisition of an isothermal remanent magnetization (IRM) was performed by exposing specimens to increasing magnetic fields up to 0.65 T at room temperature. The IRM was thermally demagnetized following the same procedures for thermal demagnetization of the NRM [e.g., *Lowrie and Heller*, 1982]. Magnetic susceptibility was calculated using a Sapphire Instruments SI-2 sensitive to about 10^{-8} SI volume units.

Hysteresis loops for representative samples were acquired with an alternating force magnetometer (AGFM) at the University of Utah. For each hysteresis run the coercivity H_c, saturation magnetization M_s, and saturation remanent magnetization M_{rs} were determined. Backfield demagnetization was also performed which allows a determination of coercivity of remanence H_{cr}.

Petrographic studies were conducted on the various carbonate lithologies using reflected and transmitted light. Magnetic mineralogy was studied on the scanning transmission electron microscope (STEM) by D. Suk at the University of Michigan. Strontium $^{87}Sr/^{86}Sr$ and oxygen stable isotopes were measured by G. Gao at the University of Texas.

Paleomagnetic Results and Interpretations

Fourteen of the 18 carbonate sites sampled carry a well-defined, stable characteristic remanence and were used in a paleomagnetic fold test. Clastic lithologies had weak magnetizations, and the decay was poorly defined with no consistent directions.

Several magnetic components were found in Belden Formation carbonates. At low demagnetization temperatures (<200°C) a northerly and steep postfolding component (Figure 3) was removed from all specimens. At intermediate temperatures (200-400°C) a component with southerly declinations and negative inclinations was removed. All carbonate beds sampled (18) on the fold, having sufficient NRM intensity to be measured (16), carry this reversed component. In several beds this was the only component found. A fold test on this component indicates that it is postfolding (Table 1b); the grouping before tilt correction is statistically significant at the 99% level using the *McElhinny* [1964] test. Before structural correction the mean declination for this component before is 186.7°E and the mean inclination is -54.6° (precision parameter (k)=44.8, radius of 95% probability cone ($\propto_{95}$)=4.9°). The results for this pervasive component suggest remanence acquisition in the Tertiary.

A characteristic component with northerly declinations and moderate-to-steep inclinations (Figure 3) was removed at high temperatures (400-580°C) and is found in the majority of carbonate beds. The intensity of this component varies between sites by several orders of magnitude (9.0×10^{-4} to 9.0×10^{-2} A/m). Applying the tilt correction suggests that this characteristic magnetization is synfolding (Figures 4a and 4b and Table 1).

Site directions from the steep westerly limb and the easterly limbs were evaluated using the fold test proposed by *McFadden and Jones* [1981]. This test distinguishes whether the mean direction of a group of sites from one limb can be statistically distinguished from the mean direction of sites from another limb. For the Belden Formation the hypothesis of a true common mean in the McFadden and Jones formulation of the fold test can be accepted at the 95% confidence level for the 58-72% unfolding steps for the characteristic magnetization (Figure 5). It must be rejected for the mean directions before and after tilt correction. The *McElhinny* [1964] test shows a similar result with a best grouping (k) at 65% unfolding (Figure 5). At 65% unfolding the mean declination is 338.7°E, the mean inclination is 62.5° (k= 54.9; $\propto$95= 5.9°).

Table 1a. Magnetization Directions and Statistics for Magnetic High-Temperature Component

	Site Means										
			Before Structural Correction					After Structural Correction			
	Strike deg	Dip, deg	*n/n0*	*D*, deg	*I*, deg	*k*	$\propto 95$,	*n/n0*	*D*, deg	*I*, deg	*k*	$\propto 95$, deg
Steep											
AM2	170	62	5/7	32.8	38.8	28.0	14.7	5/7	321.9	49.6	28.0	14.7
SW11A	163	71	7/7	44.6	37.6	79.8	6.8	7/7	299.2	57.8	82.7	6.7
SW2S	149	87	6/6	46.2	46.1	58.8	8.8	6/6	252.3	48.2	93.6	6.9
SW11	163	71	5/7	43.1	44.1	1527.0	11.9	5/7	291.5	54.7	1656.0	1.9
Shallow												
SW3	35	14	5/8	329.6	51.3	84.6	7.7	5/8	359.7	62.5	98.7	7.7
SAM2	0	15	6/7	321.0	57.6	64.0	8.4	6/7	347.0	65.0	52.6	10.6
SW11C	0	15	5/6	329.9	59.9	108.3	7.4	5/6	356.6	65.8	109.6	7.3
SAMA1	0	15	8/10	341.5	58.6	50.2	7.9	8/10	6.5	62.2	66.5	6.8
Hinge												
SW10	38	15	5/6	4.6	44.6	198.9	6.5	5/6	19.7	51.2	198.8	6.5
TAM1	349	49	6/6	289.8	40.6	33.3	11.8	6/6	358.3	66.8	33.3	11.8
SW11B	359	26	7/7	328.0	47.7	30.3	11.1	7/7	1.3	55.8	30.3	11.1
TAM2	349	49	4/5	306.8	31.6	342.8	2.9	4/5	350.8	50.8	392.9	4.6
TAMA2	349	49	7/7	300.0	40.2	27.1	11.8	7/7	11.2	56.5	25.4	12.2

Sites are grouped by structure position on the fold. Strike and dip are for the bedding at sample sites. The *n/n0* is the ratio of samples containing a stable component to samples analyzed. *D* is declination, *I* is inclination, *k* is the precision parameter associated with the site mean [*Fisher*, 1953], and $\propto 95$, is the radius of a 95% probability cone for the site mean.

Table 1b. Magnetization Directions and Statistics for Magnetic Characteristics and Intermediate-Temperature Reversed Components

Means of Sites	*D*, deg	*I*, deg	*n/n0*	*k*,	$\propto 95$, deg	Pole Position Longitude, deg	Pole Position Latitude, deg	*dp*	*dm*
				Characteristic Component					
Before structural correction	345.7	54.7	12/13	7.2	17.4	145.6E	77.7N	17.4	24.6
After structural correction	335.7	63.2	12/13	16.4	11.1	173.8W	71.5N	13.7	17.4
65% Tilt correction	338.7	62.5	12/13	54.9	5.9	175.6W	73.6N	7.2	9.2
Steep Flank	334.1	68.7	4/4	30.9	16.8				
Shallow Flank	346.2	61.8	4/4	292.3	5.4				
Hinge	335.0	56.7	4/5	79.9	10.3				
				Intermediate Temperature Reversed Component					
Before structural correction	186.7	-54.6	20/25	44.8	4.9	21.9E*	82.9N*	4.9	6.9
After structural correction	184.7	-48.7	20/25	3.8	18.7				

See Table 1a for definitions. Twenty-five samples from across the fold were analyzed for the intermediate reversed component.
The dp and dm are semiaxes of the 95% cone of uncertainty around the pole.

*These values denote antipodal pole.

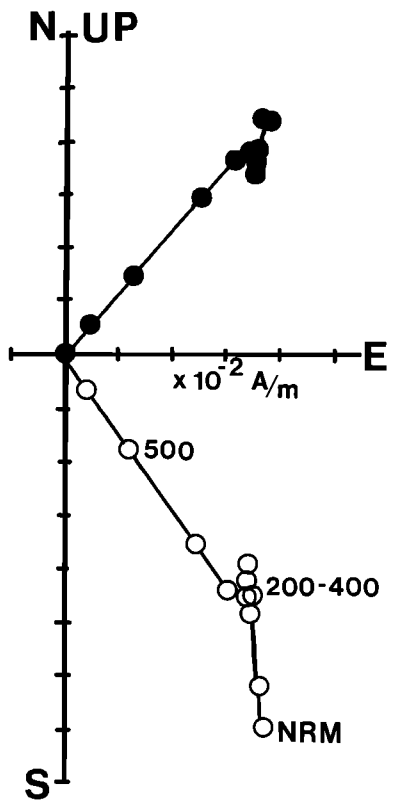


Figure 3. *Zijderveld* [1967] diagram of sample from SW11 bed (steep flank, before tilt correction) showing the following three components: northerly and down (natural remnant magnetization-200°C), southerly and up (200-400°C) and the characteristic northerly and down (400-550°C). Solid circles are horizontal component; open circles, vertical component.

The pole position for the characteristic component falls near the Cretaceous-early Tertiary part of the North American apparent polar wander path (Figure 6), consistent with timing for the initiation of the White River Uplift [e.g., *Nuccio et al.*, 1989] and the folding at Sweetwater.

Acquisition of a saturation isothermal remanent magnetization (SIRM) and thermal demagnetization of the SIRM for representative samples suggest the probable carrier of the remanence is magnetite. Representative samples show saturation by 0.2 T and complete removal by 580°C in most cases. This is consistent with demagnetization results which indicate maximum unblocking temperatures of less than 580°C and removal of the component by 100 mT during AF treatment. Minor hematite is indicated by small increases in IRM above 0.2 T and a small magnetization remaining above 580°C in some specimens.

Hysteresis properties were investigated to determine characteristics of the magnetite. Samples of the SW11 bed which carries the characteristic component have wasp-waisted hysteresis loops (Figure 7a). This suggests a bimodal distribution of magnetite coercivities (grain sizes). The values for H_{cr}/H_c (mean = 2.14; standard deviation = 0.19; number of samples n=3) and M_{rs}/M_s (mean = 0.381; standard deviation = 0.015; n=3) are high when compared to values for a synthetic mixture of the single domain (SD) plus multidomain (MD) grains from *Parry* [1982] but are consistent with the results from other remagnetized carbonates [*Jackson*, 1990; *McCabe and Channell*, 1984]. These results suggest that the limestones with the characteristic component contain a mixture of SD and MD or superparamagnetic magnetite [*Jackson*, 1990; *Jackson et al.*, 1993; *Channell and McCabe*, 1994]. The SD magnetite is interpreted to carry the characteristic remanence [e.g., *Jackson*, 1990].

In contrast, samples which only contain the low-temperature northerly and intermediate-temperature reversed components have different hysteresis properties (Figure 7b). H_{cr}/H_c values which average 4.41 (standard deviation = 0.014; n= 4) and M_{rs}/M_s values which average 0.109 (standard deviation = 0.014; n= 4) plot close to values reported by *Parry* [1982] for a mixture of SD and MD grains. These results suggest that the magnetite in the C bed is probably coarser (e.g., pseudo single domain and multidomain) than in the limestone with the characteristic component.

The northerly and steep postfolding component removed at low demagnetization temperatures is interpreted as a modern viscous magnetization. The postfolding component with southerly declinations that is removed at intermediate temperatures (200-400°C) apparently resides in multidomain magnetite and is interpreted as a thermoviscous remanent magnetization (TVRM). Laboratory unblocking temperatures (maximum 400°C) are consistent with acquisition of this component during cooling from the modeled burial temperatures of 150°C [*Nuccio et al.*, 1989] based on experimental evidence presented by Kent [1985] which assumes the presence of multidomain magnetite.

On the basis of the blocking temperature-relaxation time curves from *Pullaiah et al.* [1975], the low burial temperatures for the Belden Formation and the 550-580°C unblocking temperatures suggest that the characteristic component is not a TVRM. The origin of this component is a crucial issue. Petrographic and geochemical studies, described in the following sections, were undertaken to assess this issue.

Strontium Isotopes, Vein Test, and Magnetic Characteristics versus Structural Position

Measurement of $^{87}Sr/^{86}Sr$ values in vein material and host rock was undertaken to determine the nature of the mineralizing fluids (e.g., basinal or meteoric) and to test for a link between mineralizing synfolding (orogenic) fluids and remagnetization. In addition, individual beds in highly mineralized areas and relatively unmineralized areas were sampled.

The $^{87}Sr/^{86}Sr$ values of 0.70880 to 0.70985 (Figure 8) for the limestones at the Sweetwater locality are radiogenic relative to Pennsylvanian coeval seawater values of 0.70820-0.70850 [*Burke et al.*, 1982]. Basinal fluids (e.g., *Gao et al.* [1992], and many others) as well as orogenic fluids are likely to be radiogenic. These results are consistent with a working hypothesis that alteration by the synfolding mineralizing fluids caused the characteristic magnetization.

A transect of samples across a calcite vein and the surrounding packstone indicates an isotopic halo, with the vein material being the most radiogenic (Figure 9). The $^{87}Sr/^{86}Sr$ values decrease away from the vein in both directions. The mineralizing fluids which entered the fractures were therefore responsible for the observed isotopic alteration in the packstone. Both magnetic susceptibility (Figure 9) and NRM intensity, however, show no apparent influence by the

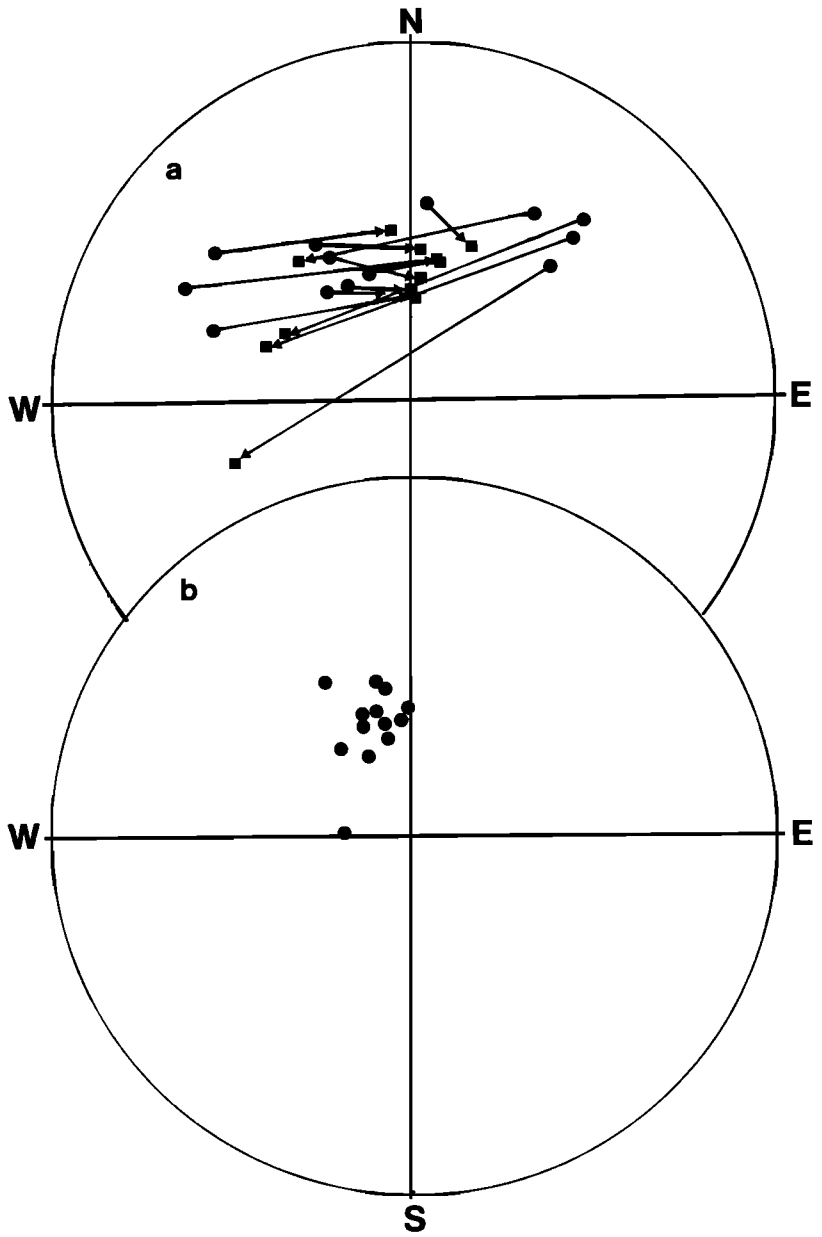


Figure 4. (a) Equal-area projection of both before (circles) and after (squares) tilt mean site directions for the characteristic component. Directions cross during unfolding. (b) Best grouping of site means at 65% unfolding.

proximity to the vein. There is no magnetic halo accompanying the isotopic halo. The variation in the magnetic values is not related to the veins but appears to be related to an increase in grain size from the base to the top of the sampled bed.

Comparison of NRM intensities and intensities of the characteristic magnetization from different structural positions on several individual beds yields no consistent trend. Overall, highly mineralized areas have no higher NRM intensity, characteristic magnetization intensity, or magnetic susceptibility than the relatively unmineralized sites on the shallow flank.

The influence of lithology on magnetic intensity becomes apparent when sample sites across the fold are sorted on the basis of NRM intensity and lithology (Figure 10). In general, packstones have the highest intensity, whereas mudstones and wackestones are weaker and show no trend relative to each other.

Petrography, Scanning Transmission Microscopy, and Selective Destructive Demagnetization

Petrographic examination shows that magnetically strong beds contain dark spheroidal or elliptical fecal pellets (Figure 11) which are primary. The pellets are irregularly distributed and in some cases are associated with burrowing. These pellets commonly contain clay minerals, organic material, and pyrite grains with alteration rims of iron oxides.

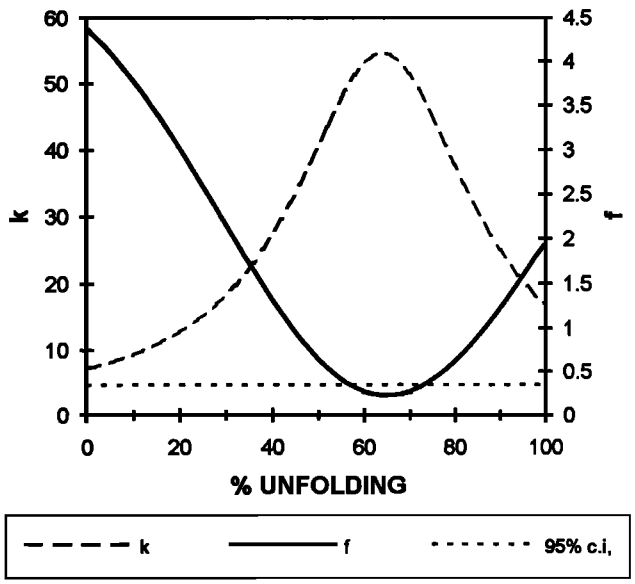


Figure 5. Incremental fold test. Values of the f [*McFadden and Jones* 1981] fold test statistic and precision parameter k [*McElhinny*, 1964] are plotted against percentage unfolding. Maximum k is at 65% unfolding, whereas minimum observed f is at 65% unfolding. The 95% confidence limit is shown for the McFadden and Jones [1981] test.

Certain echinoderm fragments in the packstones also show partial replacement by iron oxides. In contrast, pyrite found in clean limestone is typically unaltered.

In order to identify the oxide phases in the reaction rims around pyrite in the pellets, thin sections were prepared and examined at the University of Michigan by D. Suk on the Phillips CM-12 STEM equipped with a Kevex quantum energy dispersive analysis (EDA) system. SEM examination of the opaque grains within the pellets show them to be composed of two distinct phases, a brightly reflective interior and duller replacement rims (Figure 12). EDA analysis of the interiors shows both Fe and S, whereas the rims contain dominantly Fe. Electron diffraction patterns of the interiors indicate pyrite, whereas the patterns for the rim material are diagnostic of magnetite (Figure 13). Alteration of pyrite to magnetite is interpreted as the mechanism for magnetite authigenesis and perhaps remanence acquisition. Other workers have documented that magnetite can replace pyrite in carbonates [e.g., *Suk et al.*, 1990].

The identification of magnetite in the pellets and the observation that rocks with many pellets have the strongest characteristic magnetization is suggestive but not conclusive proof that this magnetite is the carrier of the remanence, rather than a dispersed magnetite in the matrix. By removing the pellets from a sample with a known magnetization, it may be determined if the magnetite identified in the pellets does carry the remanence. The strength of the NRM ($9x10^{-2}$ A/m) in the SW11 packstone bed makes it possible to measure the magnetic vector in a single-standard thin

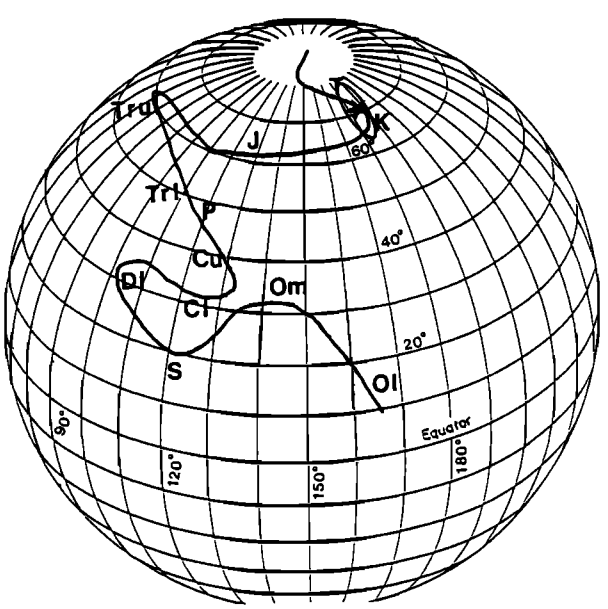
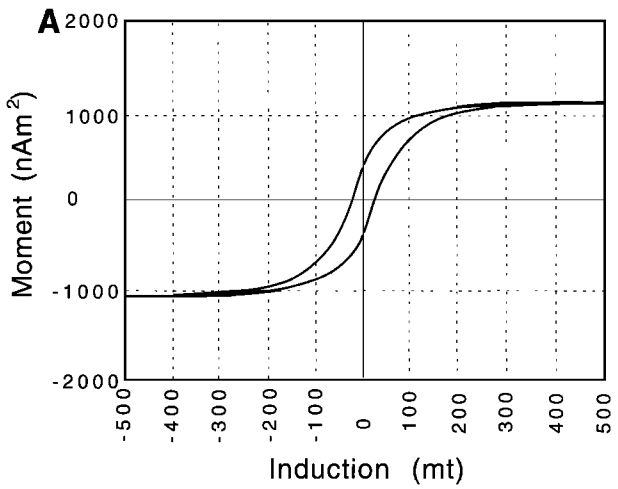


Figure 6. North American polar wander path for lower Ordovician through Modern [*Van der Voo*, 1989]. Star with 95% uncertainty cone shows pole position for the 65% unfolding direction.



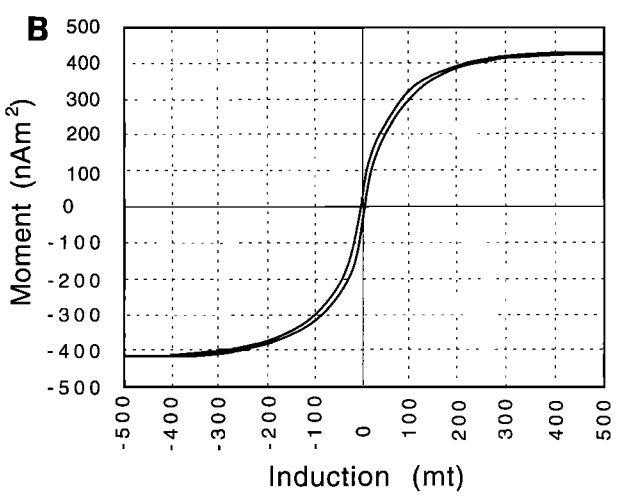


Figure 7. Representative hysteresis loops for (a) limestone beds with the characteristic component (SW11 bed) and (b) samples from the C bed which contains the northerly component removed at low temperatures and the reversed component removed at intermediate temperatures but not the characteristic component.

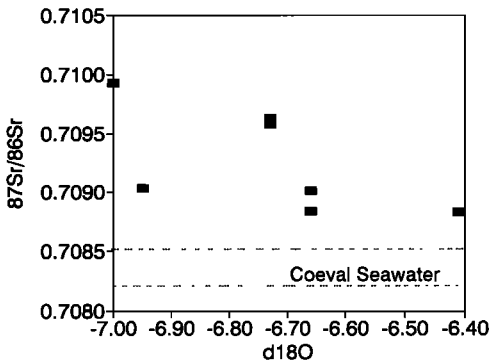


Figure 8. The $^{87}Sr/^{86}Sr$ versus $\delta^{18}O$ for Belden Formation carbonates in the study area. Values for Pennsylvanian coeval seawater (Pee Dee belemnite) are from *Burke et al.* [1982].

section [e.g., *Larsen*, 1981]. To increase the signal to noise ratio, a series of 10 2.5-cm-diameter, oriented, disk-shaped thin sections were cut from a single core from the SW11 bed. All of the pellets in the thin sections were identified and marked. The oriented sections were stacked into their original configuration and measured in the magnetometer. In each subsequent iteration all pellets from one disc were removed with a diamond scribe under a binocular microscope. The individual disk was measured before and after the removal of the pellets, and then the stack was reassembled and remeasured.

The results can be evaluated by first examining thermal demagnetization results for a sample from the SW11 bed (Figure 14). At low temperature (<200°C) the magnetic intensity drops as the modern viscous component is removed. At higher temperatures (200°C to 375°C) as the southerly TVRM is removed the magnetic intensity actually increases because the direction of this component is opposed to the stronger characteristic magnetization. Finally, at temperatures greater than 375°C the intensity falls again as the characteristic magnetization is removed.

In the selective demagnetization, intensity drops rapidly with the removal of the pellets (Figure 15). With all the

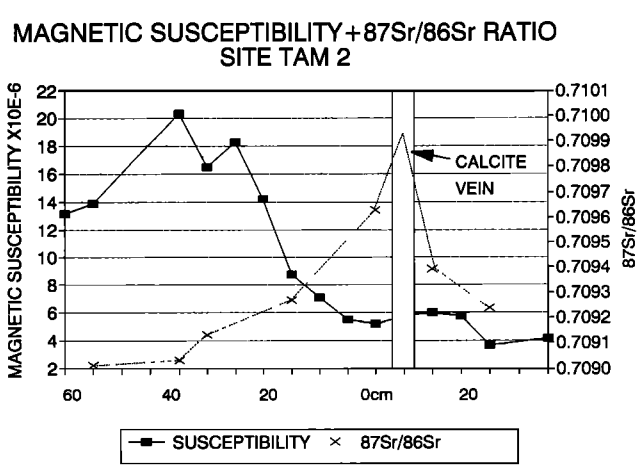


Figure 9. Results of vein test showing a marked strontium isotope halo surrounding the calcite vein. The vein cuts a packstone which coarsens upward. Up is to the left. Magnetic susceptibility (SI volume units) variations show no relation to vein proximity.

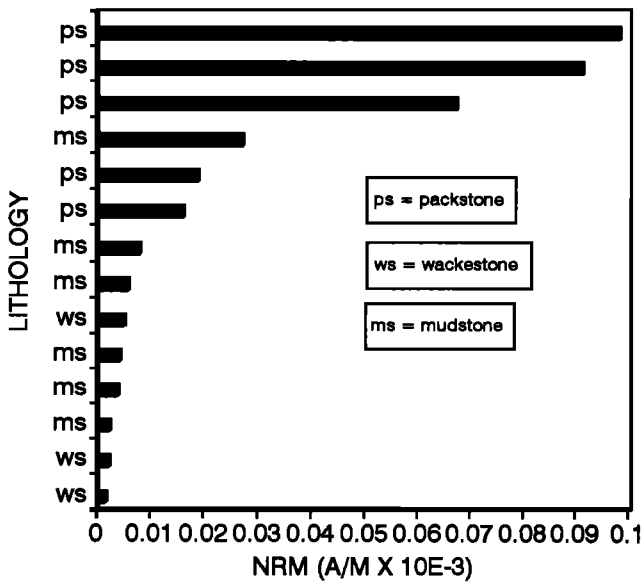


Figure 10. Bar graph illustrating that NRM intensity for carbonate sites shows a strong relation to lithology, with packstones having considerably stronger remanence than carbonate mudstones and wackestones.

pellets removed from 7 of the 10 thin sections only about 10% of the area of the thin sections has been removed but the magnetic intensity has decreased 75%. Intensity begins to increase after step 7 as the characteristic component, carried preferentially in the pellets, has been diminished by pellet removal to the level of the opposing southerly TVRM which is further unmasked as the remaining pellets are removed (Figure 16).

This test indicates that the characteristic magnetization is dominantly carried in the pellets. This result is also consistent with the observation that beds with many pellets have a strong characteristic magnetization, whereas beds with few or no pellets have a weak characteristic magnetization, or only the southerly component and a modern overprint. Because the characteristic magnetization has been localized in the pellets and the carrier is likely to be SD magnetite, either the iron oxides rimming the pyrite grains are composites made up of a mass of fine particles or there is additional fine magnetite in the body of the pellets.

The TVRM is present in the matrix and may also be present in the pellets. Iron oxide spheres similar to magnetite reported from other limestones [e.g., *McCabe et al.*, 1983] are found in the matrix of limestones that contain the characteristic magnetization as well as in limestones that only contain the TVRM and the modern viscous component. These spheres, interpreted to be magnetite, may carry the TVRM. The distribution of the magnetite carrying the modern viscous component is unknown; it could be present in both the matrix and the pellets.

One black carbonate mudstone (site SW2S) with a strong magnetization similar to that in the some packstones (Figure 10) either has pellets which cannot be distinguished in the dark matrix or has a matrix composition of clay minerals and/or organic matter similar to that in the pellets. Some beds within the Belden Formation that did not show the characteristic component contain pyrite without alteration rims in a relatively clean limestone matrix.

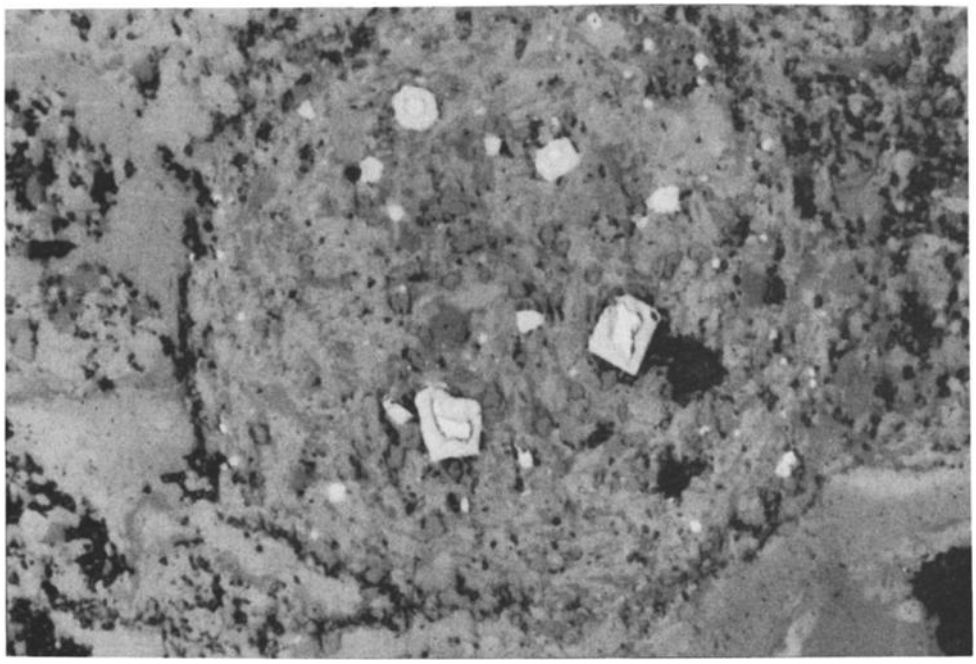


Figure 11. Photomicrograph of a fecal pellet (reflected light). Fecal pellets generally contain pyrite rimmed with iron oxides. Pellet is 250u m in diameter.

Discussion

The petrographic evidence for authigenic magnetite and the low burial temperatures, which rule out a thermoviscous remagnetization mechanism, suggest that the synfolding characteristic magnetization in the Belden Formation is a CRM. When considering the origin of this magnetization, several issues need to be addressed. For example, there are a number of mechanisms [e.g., *Hudson et al.*, 1989] that can produce a synfolding test result, and these mechanisms need to be evaluated for the origin of the characteristic magnetization. The chemical mechanism of magnetite authigenesis also needs to be addressed.

In addition to remanence acquisition during folding, a number of other mechanisms can produce a synfolding test result [e.g., *Hudson et al.*, 1989]. These include structural modifications of a prefolding magnetization and combinations of prefolding and postfolding (or synfolding) components.

In terms of remanence acquisition during folding, the most popular model for magnetizations (e.g., low burial temperatures, presence of authigenic magnetite) similar to the characteristic magnetization in the Belden Formation has been a chemical process triggered by orogenic fluids [*McCabe et al.*, 1983; *Oliver*, 1986; *Jackson et al.*, 1988; *McCabe et al.*, 1989; *McCabe and Elmore*, 1989; *Lu et al.*, 1991; *Suk et al.*, 1990; *Oliver*, 1992]. The lack of a control on the distribution and intensity of the magnetization by the veins and structural position, however, does not support this hypothesis for the characteristic magnetization in the Belden Formation. The magnetic characteristics of the Belden Formation vary with primary lithologic characteristics and do not show a relationship to alteration as documented by isotopic values or syntectonic mineralization.

An alternative explanation, however, for the lack of a link between syntectonic mineralization/fluid access and magnetic intensity is that matrix permeability may have exerted a control on the distribution of fluids and thereby the remagnetization. For example, *Grant et al.* [1991] show that permeability is controlled by the stratigraphy of carbonate parasequences. Relative matrix permeability to fluids as a control for a synfolding magnetization in the Belden Formation is considered unlikely for several reasons. For



Figure 12. Scanning electron photomicrograph of pyrite grain with iron oxide rim from fecal pellet.

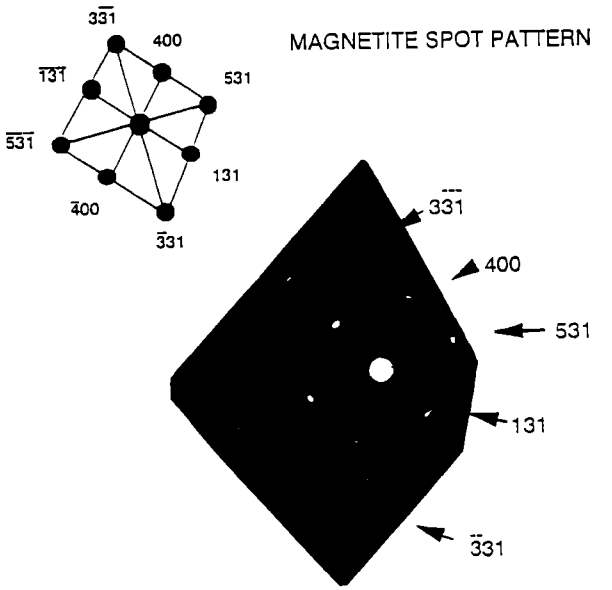


Figure 13. Diffraction patterns for the rimming oxides taken on the Scanning transmission electron microscope showing cubic faces of magnetite.

example, the black carbonate mudstone bed with high intensities in the range of the packstones probably had very low matrix permeability. More importantly, the permeability of the Belden Formation was changed with the onset of deformation. Shattered mudstones in the hinge area had considerably higher permeability after deformation than relatively unfractured packstones on the shallow limb, yet the packstones have higher magnetic intensities.

The characteristic magnetization in the Belden Formation may be a CRM, but there is no evidence that it is related to syndeformational fluids. Other mechanisms that have been proposed for true synfolding magnetizations include stress-accelerated acquisition of a VRM and acquisition of a piezoremanent magnetization (PRM) [e.g., *Hudson et al.*, 1989]. Experimental studies suggest that geologically reasonable differential stress levels can cause acquisition of a PRM [e.g., *Borradaile*, 1994]. Other experimental studies, however, indicate that stress-induced changes are most likely in MD grains and in Ti-rich grains [e.g., *Kean et al.*, 1976]. A PRM therefore might be a valid explanation in a clastic unit, such as the Jurassic Preuss Sandstone, where the magnetization apparently resides in detrital magnetite [*Hudson et al.*, 1989], but would seem less likely in the Belden Formation, where the characteristic magnetization resides in single-domain magnetite. A PRM that resides in MD grains could also overprint the characteristic component.

Another argument against the stress-related mechanisms is that stress fields during folding were probably complex, yet during unfolding the sites from different areas on the fold all behave in a similar fashion. For example, sites from the steep flank show similar amounts of oversteepening. On the basis of the different fracture patterns and variation in the degree of fracturing, the magnitude of stress and the orientation of the principal components of stress varied between sites and changed during development of the fold. If stress had caused acquisition of a PRM, a less uniform synfolding test result might be expected.

Several different models have been proposed for structural modification of a prefolding remanence into a synfolding configuration [e.g., *van der Pluijm*, 1987; *Kodama*, 1988; *Stamatakos and Kodama*, 1991]. If the geometry of the strain is correct to have rotated the magnetization into a synfolding configuration, a syntectonic remanence acquisition must be questioned [*Kodama*, 1988]. Only if the deformation is sufficiently understood to determine that the difference between a synfolding and prefolding remanence cannot be accounted for by strain or rigid body rotation can a definitive determination of a synfolding magnetization be made [*van der Pluijm*, 1987; *Stamatakos and Kodama*, 1991].

The characteristic component for the Sweetwater samples is dispersed along a great circle normal to the fold axis after tilt correction. The inferred sense of rotation of the remanence on the steep limb is opposite to the rotation of the beds. If the remanence on the steep flank had a similar orientation to bedding as the shallow flank remanence (subnormal to the bedding surface), the directions would correct to a best grouping after tilt correction. It is possible that strain could account for the synfolding fold test. The question is therefore can it be established that the steep flank characteristic magnetization has been steepened relative to bedding during deformation?

As one means to evaluate relative strain in the fold, fracture counts were taken for two lithologies: thinly bedded (5-8 cm) carbonate mudstone and a 50-cm thick packstone bed. Three meter counts along bedding strike were taken for each lithology in the steep flank, near the hinge and on the shallow flank. The values for the packstone are 36 (steep), 29 (hinge), and 13 (shallow), whereas the values for the mudstone are 570 (steep), 1050 (hinge), and 240 (shallow). Clearly, both lithology and structural position affect fracture intensity. The fine-grained and thinly bedded mudstones in the steep flank and hinge show the greatest intensity of fracturing, and the coarse-grained thick packstone bed on the shallow flank shows the least. Most fractures on the shallow flank show little mineralization, whereas mineralized fractures are common in the other two locations. Interpenetration of grains and stylolites are also more highly developed in the steep flank samples than in the shallow flank samples. There is also a noticeably greater degree of calcite

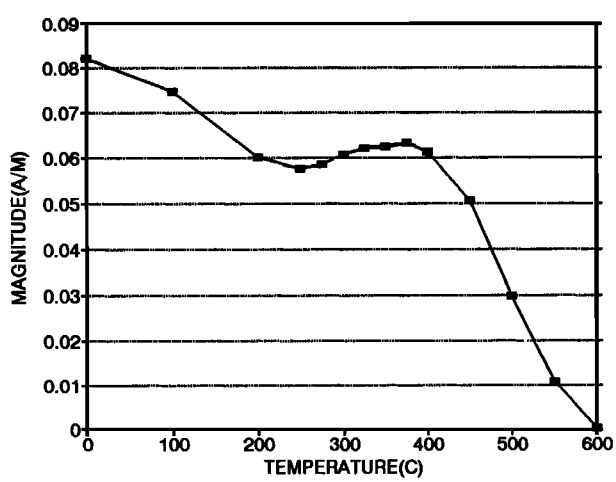


Figure 14. Thermal demagnetization of packstone sample SW11.

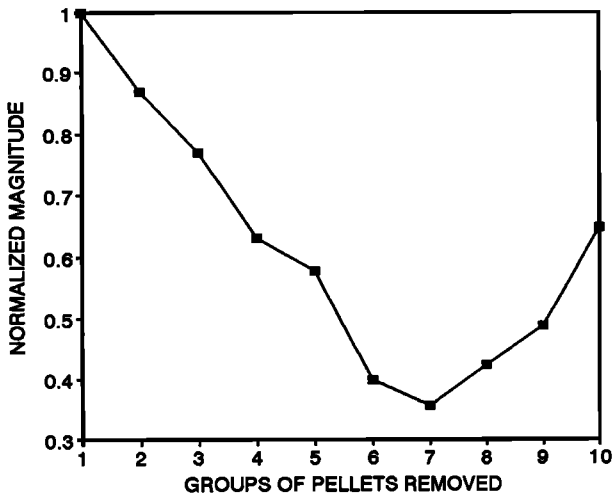


Figure 15. Graph of selective demagnetization (SW11). Groups of pellets removed correspond to steps in Figure 16.

twinning in the steep flank samples than in the shallow flank samples.

The precision of grouping of sites in each flank can be used to help distinguish a true synfolding remanence from a strain-deflected prefolding remanence [e.g. *Kodama*, 1988]. At the Sweetwater locality the blocking temperature spectra, coercivity spectra, and relative intensity from bed to bed for the characteristic component are the same on the steep and shallow flanks. It is reasonable to assume that the remanence resulted contemporaneously from the same event across the fold and therefore would originally have had similar precisions. In the incremental fold test for the Belden Formation, however, the k values for the precisions of sites on the hinge and flank positions are dissimilar in situ (25, 175, 210) at the maximum grouping (65% unfolding) for all sites (80, 30, 302) and after complete structural correction (100, 20, 350). *McFadden and Jones* [1981] note that for their fold test to be valid the directions from various limbs must originally have been part of the same population and should therefore have a common precision. The precisions of the site means from the two limbs and hinge area for the characteristic magnetization in the Belden Formation, however, clearly do not meet this criterion.

This analysis suggests the possibility that a prefolding CRM was altered into a synfolding CRM by deformation. The steep flank has higher amounts of strain than the shallow flank, but the characteristic magnetization on both flanks may have been influenced by deformation. It should be pointed out, however, that studies which have demonstrated structural modification of a prefolding magnetization have found that the deflection of the magnetic directions occurs in discrete zones that have undergone relatively high strain [e.g., *Stamatakos and Kodama*, 1991]. The Belden Formation limestones sampled are not discrete zones that have undergone high strain. It is also not clear that strain levels in the Belden were high enough to have caused deflection of a prefolding component into a synfolding configuration. In addition, sites from different lithologies on the fold behave similarly and show similar amounts of oversteepening. If strain was an important factor, the different lithologies might be expected to behave in a less uniform way because the mechanisms of strain were different in the different lithologies (i.e., shattering in mudstones and twinning and pressure solution in packstones).

There are a number of different combinations of components that could produce a synfolding test result. There is no evidence for detrital magnetite in the Belden Formation limestones, so it is unlikely that the magnetization is a combination of a prefolding detrital magnetization and a postfolding (or synfolding) component. A combination of a prefolding CRM and a postfolding (or synfolding) overprint is possible. As pointed out by *Dunlop* [1983] and *Xu and Dunlop* [1994], complete thermal demagnetization of a magnetization (i.e., modern viscous) carried in multidomain magnetite will occur only at temperatures near the Curie temperature. The characteristic magnetization in the Belden Formation could be overprinted by a modern viscous component that extends to high demagnetization temperatures. A prefolding magnetization overprinted by such a postfolding component could produce the synfolding test result. For example, if the characteristic magnetization in the Belden is overprinted by a northerly and steep-down modern component, the before-tilt directions, particularly from the steep flank, would be steepened and this could account for the synfolding test result. A prefolding CRM overprinted by a synfolding PRM could also produce the same oversteepening.

If a postfolding or synfolding overprint is present, the demagnetization results could provide some evidence for overlap between components. Examination of the demagnetization paths did not reveal any evidence of overlap; the characteristic magnetization decays in a linear fashion without any curved demagnetization paths. This would suggest that if an overprint is present at high demagnetization temperatures, it almost completely overlaps with the character-

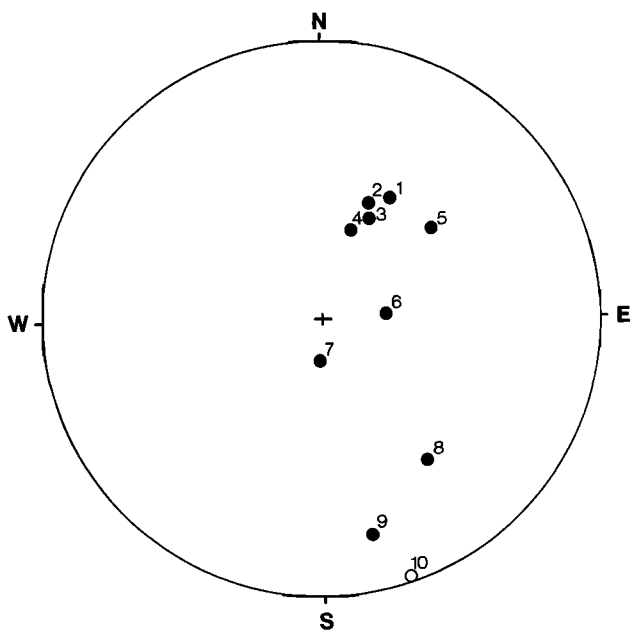


Figure 16. Equal-area stereonet projection showing the change in direction of the resultant magnetic vector as pellets are removed. Vectors change from northeasterly to southerly. Steps in this figure correspond to steps in Figure 15. Solid circles are on the lower hemisphere, open circle is on the upper hemisphere.

istic component. Although this might be considered unusual, some magnetostratigraphic studies have reported characteristic magnetizations that have a systematic bias in the directions which is interpreted to be the result of overlap with a modern viscous magnetization [e.g., *Claire et al.*, 1991]. It is not clear, however, that the magnetic characteristics (e.g., domain states, Ti content) in such cases are similar to those in the Belden Formation.

It is not possible to unequivocally establish whether the characteristic magnetization in the Belden Formation is prefolding or truly synfolding. Although a deflection by strain, a PRM, or a stress-accelerated VRM may be unlikely, they cannot be ruled out. A prefolding CRM, overprinted by a modern VRM or perhaps a synfolding PRM, are also possibilities.

Although the mechanism of remanence acquisition may be open to speculation, the original magnetization was a CRM, and there is no evidence that this CRM was related to the syndeformational mineralizing fluids. This observation is consistent with several other studies which report CRMs in limestones that have not been altered by externally derived fluids (other than meteoric fluids). For example, *Elmore et al.* [1993a] report that in the low-permeability Viola Limestone (Ordovician) in Oklahoma a Permian CRM in hematite is limited to a halo around mineralized fractures, as was an isotopic alteration signature by basinal fluids. In contrast, away from the veins the organic-rich Viola Limestone carries a pervasive Pennsylvanian synfolding CRM in magnetite and there is no isotopic evidence of alteration by basinal fluids. In addition, *Plaster-Kirk et al.* [1995] report that organic-rich lacustrine source rocks in the Devonian Old Red Sandstone in Scotland which have not been altered by an orogenic or basinal fluid contain a Carboniferous CRM residing in magnetite.

The chemical mechanism responsible for magnetite authigenesis (and perhaps remanence acquisition) in the Belden Formation is also open to question and speculation. A prefolding oxidizing fluid, perhaps meteoric in origin, is one possibility. If such a meteoric fluid did enter the rock, it did not leave an obvious diagnostic signature that can be detected geochemically. An alternative model involves replacement of pyrite during burial and maturation of hydrocarbons. In this model, ferric iron that is complexed by humic ligands [e.g., *Fontes et al.*, 1992] is released during maturation and triggers the oxidation of pyrite into magnetite. This reaction has been successfully modeled in simulation experiments which correspond chemically to the equilibration of pyrite in an organically encased, acidic microenvironment (e.g., fecal pellets) surrounded by basic fluids in a carbonate rock [*Brothers et al.*, 1994]. This model is also consistent with other studies which report an association between organic matter and authigenic magnetite [e.g., *Elmore and Crawford*, 1990; *Elmore et al.*, 1993b].

Conclusions

The Belden Formation carbonates at the Sweetwater Creek Anticline were evaluated for a link between mineralizing (orogenic) fluids and an apparently synfolding remanence. The characteristic remanence for the Belden Formation carbonates shows a crossing pattern upon stepwise unfolding and superficially could be linked to radiogenic syntectonic mineralizing fluids. Testing this hypothesis, however, does not reveal a connection between the extent of isotopic alteration or mineralization related to syntectonic fluids and the presence or intensity of the characteristic remanence. In particular, there is no relation between a geochemical halo and the magnetization around veins. Furthermore, the same bed does not have a consistently higher characteristic magnetization when measured on the highly mineralized steep flank relative to the less mineralized shallow flank. The characteristic magnetization in the Belden Formation is lithologically controlled in its distribution. The beds with the highest magnetization have concentrations of pyrite, with alteration rims of magnetite in fecal pellets containing clay minerals and organic material. Selective demagnetization indicates that the pellets contain the magnetization. Rock magnetic studies suggest the characteristic magnetization resides in SD magnetite.

It is not possible to unequivocally establish whether the characteristic magnetization is prefolding or synfolding. We can rule out a true synfolding component that was caused by the syndeformational fluids. A synfolding stress-related magnetization, such as a PRM or a structural modification of a prefolding component, are also possible, but they are not considered likely. A prefolding component overprinted by a modern VRM in multidomain grains is one possible, and perhaps likely, explanation for the synfolding test result.

The characteristic magnetization in the Belden was originally a CRM, and remanence acquisition probably took place during burial, prior to deformation. The chemical mechanism responsible for magnetite authigenesis is unknown but could be a prefolding oxidizing fluid, or perhaps processes associated with maturation of hydrocarbons are possibilities.

Acknowledgments. This work was supported by NSF grant EAR9218960 and DOE grant DE-FG02-93ER14397. We thank G. Gao and D. Suk for running the analyses and K. Kodama, D. Stearns, M. Engel, D. London, and J. Furneaux for useful reviews. Tony Hillegeist and Mark Phillips assisted in the field work.

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(Received June 13, 1994: revised November 28, 1994; accepted January 4, 1995)